

7(a)(i)	as mutation rate increases lifespan decreases ;	1
7(a)(ii)	any three from: 1 high(er) mutation rate gives more, genetic diversity / genetic variation / alleles / chance of beneficial allele ; 2 short(er) lifespan gives more generations (per year / per unit time) ; 3 with, high(er) mutation rate / short(er) lifespan, they, adapt / evolve, better / more easily / faster ; 4 mouse (and ferret) may, adapt / evolve, faster ; 5 AVP ;	3

Question	Answer	Marks
7(b)(i)	<i>comparison:</i> C1 genetic engineering faster / selective breeding slower ; <i>any one reason from:</i> R2 selective breeding occurs over (many) generations ; R3 all cows could be surrogate mothers for engineered embryos (giving large increase in allele frequency in one generation) ; R4 AVP ;	2
7(b)(ii)	<i>any two from:</i> 1 (improve named) quality of, animal / milk ; 2 (improve) quantity / yield, of milk ; 3 AVP ;	2

Question	Answer	Marks
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